

A Manicka Praveen

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EDUCATION

Singapore University of Technology and Design

Sep 2025 – Aug 2027

Master of Engineering (Research) in Computer Science and Design, **CGPA: 4.90/5.00**

Relevant Modules: Multimodal GenAI, Fundamentals of Robotics and Multi-Robot Systems, RL for Embodied AI

Nanyang Technological University, Singapore

Aug 2020 – May 2024

Bachelor of Engineering (Hons) in Mechanical Engineering, Specialisation: **Autonomous Systems**

RESEARCH PUBLICATIONS (* equal contribution)

A. M. Praveen, Y. Loo, M. B. Prasetyo, M. Meghjani

Nov 2026

"Playing to Strengths: Task Distribution for Heterogeneous Multi-Robot Teams using Monte Carlo Tree Search Guided Dialogue."

International Symposium on Robotics Research (ISRR), 2026 – Oral Presentation

Also presented at the 3rd Workshop on Semantic Reasoning and Goal Understanding in Robotics, RSS 2026

M. B. Prasetyo*, S. V. Thengane*, A. M. Praveen*, Y. Loo, M. Meghjani

Sep 2026

"Seeing Fast and Slow: Bimodal 3D Scene Graphs for Open-set Tasks." arXiv:2605.31067

Under review

S. Shrivastava, A. M. Praveen, M. Meghjani

Aug 2026

"GoGet: A Multi-Model Multimodal Assistive Agent with Disagreement-Gated Grounding."

International Conference on Advanced Algorithms and Neural Networks (AANN), 2026 – Oral Presentation

A. M. Praveen, Y. Loo, M. Meghjani

Dec 2025

"LLM-Based Coordination for Heterogeneous Multi-Robot Warehouse Exploration."

Poster presentation, Singapore Robotics Research Showcase, jointly organized with IEEE MRS 2025

V. Rao, V. T. Truong, A. M. Praveen, et al., P. Valdivia y Alvarado

Evaluating Physical Safety of Soft Grippers in Adaptive Human to Robot Handovers."

In preparation

ACADEMIC PROJECTS/COMPETITIONS

ICRA Robotic Grasping and Manipulation Competition 2025

May 2025

2nd Place - Handover Track, 3rd Place - Simulation Sorting Track

- Built real-time object pose estimation pipeline for human-to-robot handover tasks, integrating vision outputs with downstream manipulation stack, contributing to 2nd place finish

RobotX Challenge 2026

Nov 2026

- Leading a 27-member undergraduate and postgraduate team across autonomy, perception, and communications for a USV, UUV, UAV fleet

Zero-shot Object Detection and Referring Expression Comprehension

Jun 2023 – May 2024

Final Year Project

- Developed a zero-shot object detection pipeline and integrated referring expression comprehension for natural language-based object grounding within a modular ROS architecture
- Improved task success rate from 38% to 70% by integrating GPT-4V for multimodal reasoning

WORK EXPERIENCE

NAMIC, Singapore University of Technology and Design

Feb 2025 – Present

Research Assistant

- Built an end-to-end multi-camera 3D scanning system for lower-limb orthosis design in clinical settings, enabling 3D reconstruction for patient-specific workflows; filed technology disclosure as **co-inventor**
- Developed a machine learning pipeline for 3D anatomical landmark detection and automated orthosis generation using parametric design tools, bridging scanned geometry with clinical measurement requirements; filed technology disclosure as **principal inventor**
- Delivered a proof-of-concept system meeting requirements of two competitive grants, with healthy volunteer testing completed in a clinical environment towards a future clinical trial

SESTO Robotics

Apr 2024 – Jan 2025

Robot Deployment Engineer

- Deployed and configured multi-robot systems across healthcare and industrial environments and optimized robotic workflows to improve system reliability and operational efficiency

Continental Automotive

Jun 2023 – Nov 2023

Robotics Software Engineer Intern - Perception

- Designed test plans and analyzed camera/LiDAR sensor data to support perception algorithm development and software QA processes

ST Engineering

Jan 2023 – May 2023

Robotics Software Engineer Intern – Mapping & Localization

- Contributed to swarm robotics outdoor navigation system development, focusing on multi-agent coordination, path planning, and multi-sensor (IMU, GPS, LiDAR, camera) fusion for localization
- Trained and evaluated object detection models, achieving 98% accuracy on unseen data

Singapore Armed Forces

Apr 2018 – Feb 2020

Corporal First Class in 30th Battalion Singapore Combat Engineers

- Ranked in top 10% of cohort; awarded Outstanding Soldier Award

SERVICE & LEADERSHIP

Reviewer: *IEEE ICRA 2026, ISRR 2026, IEEE RA-L 2026*

Volunteer: *5th IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS), 2025*

AIESEC NTU

Local Committee Vice President, Partnership Development

Nov 2021 – Feb 2023

- Led cross-functional teams of up to 20 across multiple large-scale events and project initiatives

Graduate Teaching Assistant

Aug 2025 – Dec 2025

Mobile Robotics

- Teaching Assistant for 50.047 Mobile Robotics (Fall 2025) under Prof. Malika Meghjani

SKILLS

Computer Applications: Linux, ROS, Python, Jupyter, Solidworks, Rhino, Grasshopper

Certifications: ROS for Beginners: Basics, Motion, and OpenCV, ROS for Beginners II: Localization, Navigation and SLAM

Languages: Proficient in English and Tamil